Experiment Number: 1

Simulation and Comparative Analysis of FDMA, TDMA and CDMA: Resource Allocation, User Capacity and Mobile-Generation Context

Aim:To simulate and compare the basic resource-allocation methods used in FDMA, TDMA and CDMA, and examine how their assumed user capacity and interference behaviour differ.

Objective 4: Resource-allocation diagrams
To visualize how FDMA, TDMA and CDMA assign frequency, time and code resources to multiple users.

CODE:

```
% Experiment 1: Resource Allocation Diagrams

clear;
clc;
close all;

figure;

% Common colors for users
colors = lines(4);

% ---------------- FDMA ----------------
subplot(1,3,1);
hold on;

for u = 1:4
   rectangle('Position', [0, u-1, 1, 0.8], ...
      'FaceColor', colors(u,:), 'EdgeColor', 'k');

   text(0.5, u-0.6, sprintf('User %d', u), ...
      'HorizontalAlignment', 'center');
end

xlim([0 1]);
ylim([0 4]);
xlabel('Time');
```

```
ylabel('Frequency Channel');
title('FDMA');
set(gca, 'YTick', []);
grid on;
hold off;

% ---------------- TDMA ----------------
subplot(1,3,2);
hold on;

for u = 1:4
   rectangle('Position', [u-1, 0, 0.8, 1], ...
      'FaceColor', colors(u,:), 'EdgeColor', 'k');

   text(u-0.6, 0.5, sprintf('User %d', u), ...
      'HorizontalAlignment', 'center');
end

xlim([0 4]);
ylim([0 1]);
xlabel('Time Slot');
ylabel('Frequency Channel');
title('TDMA');
set(gca, 'YTick', []);
grid on;
hold off;

% ---------------- CDMA ----------------
subplot(1,3,3);
hold on;

for u = 1:4
   rectangle('Position', [0, 0, 1, 1], ...
      'FaceColor', colors(u,:), ...
      'FaceAlpha', 0.22, ...
      'EdgeColor', colors(u,:), ...
      'LineWidth', 1.5);
end

text(0.5, 0.62, 'Users 1-4 share', ...
```

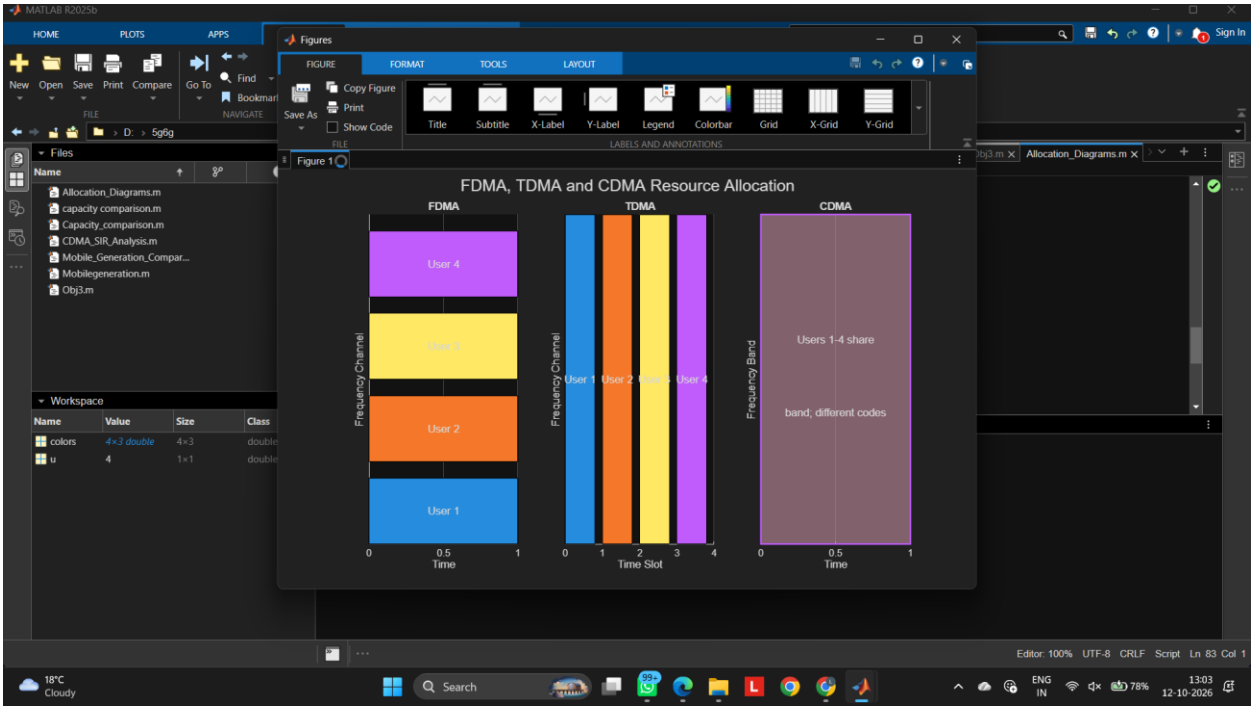
```
    'HorizontalAlignment', 'center');

text(0.5, 0.40, 'band; different codes', ...
    'HorizontalAlignment', 'center');

xlim([0 1]);
ylim([0 1]);
xlabel('Time');
ylabel('Frequency Band');
title('CDMA');
set(gca, 'YTick', []);
grid on;
hold off;

sgtitle('FDMA, TDMA and CDMA Resource Allocation');
```

OUTPUT:



ASSESSMENT RUBRICS — EXPERIMENT 1
Simulation and Comparative Analysis of FDMA, TDMA and CDMA
Performance Evaluation Rubric

Assessment Criterion	4 — Strong	3 — Meets	2 — Partial	1 — Limited
Objective 4: Resource-allocation diagrams	Clearly shows FDMA frequency separation, TDMA time slots and CDMA code-based sharing.	Diagrams understandable, with minor labeling or presentation issues.	One or more allocation concepts unclear or incorrectly shown.	Diagrams missing or substantially incorrect.
Parameter study and recorded results	Changes relevant parameters, reruns applicable programs, records outputs and explains effects.	Provides rerun results and comparison, but one change or explanation is incomplete.	Makes limited changes or records results without clear comparison.	No meaningful parameter study or usable results record.
Analysis and interpretation	Answers accurately, uses results as evidence and distinguishes model behavior from real-system behavior.	Most answers sound and refer to results, with some gaps in explanation.	Answers partly correct but mainly describe outputs without explaining them.	Analysis missing, unsupported or substantially incorrect.
Engineering judgment and conclusion	Gives a clear conclusion, identifies assumptions and limitations, and avoids unsupported performance rankings.	Conclusion is reasonable but omits a relevant assumption, limitation or implication.	Conclusion mostly repeats results, with little discussion of limitations.	Conclusion missing or draws unsupported conclusions.
Report organization and evidence	Includes code, tables, required figures, parameter values and clear captions.	Required evidence present, with minor organization or labeling issues.	Some evidence missing or difficult to follow.	Report substantially incomplete or unclear.
Individual explanation / viva	Clearly explains own work, formulas and results.	Explains most aspects, with minor gaps.	Gives partial explanations but struggles to	Cannot explain key parts of the submitted work.

connect code and
results.